

Narea Interpretation for Your Study

1. Why Narea should work (in theory)

Leaf nitrogen per area (Narea) is often used as a proxy for photosynthetic capacity because nitrogen is invested in photosynthetic machinery such as Rubisco. Therefore, higher Narea is expected to correspond to higher photosynthesis and potentially higher intrinsic water-use efficiency (iWUE).

2. What tropical studies show

Tropical forest studies indicate that while Narea and photosynthesis can co-vary along canopy light gradients, there is large interspecific variation within the same light environment where this relationship breaks down.

3. Why this decoupling happens

Narea does not capture how nitrogen is allocated within leaves, differences in mesophyll conductance, stomatal behavior, or species-specific ecological strategies. Thus, species with similar Narea can have very different photosynthetic capacities.

4. Implication for iWUE

Since iWUE integrates both photosynthesis and stomatal conductance, a weak relationship between Narea and photosynthesis across species explains why nutrient additions may change leaf chemistry without affecting iWUE.

5. Interpretation for your study

Your results are consistent with tropical forest theory: species-level physiological strategies and climatic constraints (e.g., elevation, VPD) dominate over leaf nutrient content in explaining iWUE variation.